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TITLE : QUESTIONS AND ANSWERS BASED ON RESEARCH

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Q1. What is HTTP Protocol ?

- Hypertext Transfer Protocol
- Set of rules
- Helps browser as client and a server to talk to each other on internet
- Request and send web pages , images , videos , data
- Like conversion language

Example : Restaurant

- Browser (Customer) (You)
- HTTP (Waiter)) (Carries your request)
- Kitchen (Server) (Prepares the food)
- Food (Web page) (Response)

How it works :

- Order food - Waiter takes order - Kitchen prepare - Waiter bring food
- Technically : Type in browser : http://example.com
- The browser uses the http we want and wanted server gives us the web page

Q2) What is web application ?

- Web Application
- Software program
- Runs on the internet
- Use through web browser
- No need install

Example :

- Facebook : Web Application : Post , Like , Comment
- Amazon : Web Application : Search products and buy
- No need old traditional method like newspaper or supermarket

Q3) What is webserver ?

- Computer (Software)
- Stores websites and then to browser
- Happens when we ask

Example :

- You - Customer
- Browser – Waiter
- Web Server - Kitchen
- Website - Pizza

How it works :

- Customer orders through waiter and waiter sends to kitchen and Kitchen prepares pizza
- We use the browser using web server and the web server gives us the website we want

Q4) What is HTTPS Protocol and Why is it considered secured ?

- HTTPS : Hyper Text Transfer Protocol Secure
- Secure version of HTTP
- Used when browser talks to website
- URL : http:// (Not secure) : https:// (secure)
- Considered secure with using encryption

Example :

Without HTTPS / HTTP

- Sending postcard : Anyone can read : (Postman / Stranger)
- If we write : Username : AKASH & Password : 123456
  (Anyone can read)

With HTTPS /HTTP

Example :

- Image sending locked box
- Only receiver has the key
- If stolen , they cant open
- Send : Username : Akash & Password : 123456 (Encrypted and Safe)

HTTPS (Protect)

- Login details
- Bank details
- Credit card details
- Personal data

Q5) What is Authentication and Authorization ?

Authentication :

- Checks your identity : Are you really akash

Example :

- Open Gmail
- Enter email + password
- Gmail checks and says its akash

Authorization :

- Decides what is allowed to access after authentication

Example : (What is akash allowed to do ?)

- Read our emails
- Send our emails
- Cannot read someone else emails

Technically in AWS :

Example :

- Authentication : Login using username : OTP : IAM user login
- Authorization : IAM policy : Cannot delete EC2 : Can start / stop EC2 : Cannot access S3 bucket

Q6) How does social login (like logging into some application via google or Facebook)

- No need to create new password and username each time
- Need not need the particular social media app to store password
- Login becomes faster with 1 – click

Example :

- Office Building : (Your app)
- Security Company (Google / Facebook)

How it works :

- Office trusts the security company
- If security group says this is akash
- Office lets you in
- Office never checks your ID directly

Q7) what is Synchronous and Asynchronous communication between web application ?

Synchronous Communication

Example :

- We ask - We wait - Get answer - Move on
- Sender waits until receiver responds
- Calling someone on phone : You talk - Wait - They reply

Technically :

- Web Application : We user browser to login and response comes after login and we the result based on our login in form of web page or other

Asynchronous communication

Example :

- You ask - Don’t wait - Continue work - Response later
- Sender not wait for quick response
- Sending a whatsapp message : You send - Continue work - Reply comes later

Technically :

- Web Application : Order service - Queue - Payment Later

Q8) What is REST standard or recommendation for Web applications ?

REST :

- Simple set of rules
- How web apps talk to each other over internet
- Way to design web API

Example :

In Restaurant

- Menu (API)
- You (Client) ----- Place order (Request)
- Kitchen (Server) ---- Prepares food (Processing)
- Food Served (Response)

Ideology :

- No one care how the kitchen cooks
- Care about what you get
- Therefore it is not strict standard , it is a design or practices based on recommendations